

Orientations and Manifolds with Boundary

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Orientations

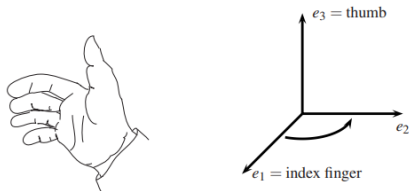


Fig. 21.3. Right-handed orientation (e_1, e_2, e_3) of $\mathbb{R}^3$.

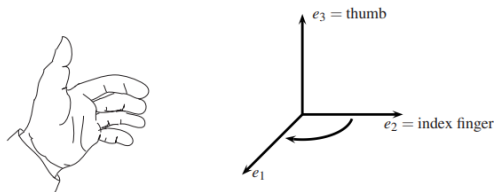


Fig. 21.4. Left-handed orientation (e_2, e_1, e_3) of $\mathbb{R}^3$.

Orientations

Equivalence relation between ordered bases of vector space

Two ordered bases

$$\mathbf{u} = [u_1 \ \cdots \ u_n] \quad \text{and} \quad \mathbf{v} = [v_1 \ \cdots \ v_n]$$

of a vector space V are said to be equivalent, written $\mathbf{u} \sim \mathbf{v}$, if

$$\mathbf{u} = \mathbf{v}A$$

for an $n \times n$ matrix A with **positive** determinant.

An **orientation** of V is an equivalence class of ordered bases.

Orientations

Equivalence relation between nonzero n -covectors

Let V be an n -dimensional vector space. We define an equivalence relation on the set of nonzero n -covectors by

$$\beta \sim \beta' \iff \beta = a\beta' \text{ for some } a > 0.$$

Since

$$\bigwedge^n (V^\vee) \simeq \mathbb{R},$$

the set of nonzero n -covectors is identified with

$$\mathbb{R} \setminus \{0\},$$

which has two connected components.

Orientations

Lemma 21.1 Let $u_1, \dots, u_n$ and $v_1, \dots, v_n$ be vectors in a vector space V . Suppose

$$u_j = \sum_{i=1}^n v_i a_j^i, \quad j = 1, \dots, n,$$

for a matrix $A = [a_j^i]$ of real numbers. If β is an n -covector on V , then

$$\beta(u_1, \dots, u_n) = (\det A) \beta(v_1, \dots, v_n).$$

Proof

$$\begin{aligned} \beta(u_1, \dots, u_n) &= \beta \left(\sum_{i_1} v_{i_1} a_1^{i_1}, \dots, \sum_{i_n} v_{i_n} a_n^{i_n} \right) = \sum_{i_1, \dots, i_n} a_1^{i_1} \cdots a_n^{i_n} \beta(v_{i_1}, \dots, v_{i_n}) \\ &= \sum_{\sigma_I \in S_n} (\operatorname{sgn} \sigma_I) a_1^{i_1} \cdots a_n^{i_n} \beta(v_1, \dots, v_n) \\ &= (\det A) \beta(v_1, \dots, v_n). \end{aligned}$$



Orientations on a Manifold

Definition

A **pointwise orientation** on a manifold M assigns to each $p \in M$ an orientation μ_p of the tangent space $T_p M$.

A pointwise orientation μ on M is said to be **continuous** at $p \in M$ if p has a neighborhood U on which μ is represented by a continuous frame; i.e., there exist continuous vector fields $Y_1, \dots, Y_n$ on U such that

$$\mu_q = [(Y_{1,q}, \dots, Y_{n,q})]$$

for all $q \in U$. The pointwise orientation μ is continuous on M if it is continuous at every point $p \in M$.

A continuous pointwise orientation on M is called an **orientation** on M . A manifold is said to be **orientable** if it has an orientation.

Orientations on a Manifold

Proposition 21.3 A connected orientable manifold M has exactly two orientations.

Proof.

Let μ and ν be two orientations on M . At any point $p \in M$, μ_p and ν_p are orientations of $T_p M$. Define a function $f: M \rightarrow \{\pm 1\}$ by

$$f(p) = \begin{cases} 1, & \text{if } \mu_p = \nu_p, \\ -1, & \text{if } \mu_p = -\nu_p. \end{cases}$$

Fix a point $p \in M$. By continuity, there exists a connected neighborhood U of p on which $\mu = [(X_1, \dots, X_n)]$, $\nu = [(Y_1, \dots, Y_n)]$ for some continuous vector fields X_i and Y_j on U . Then $Y_j = \sum_i a_j^i X_i$, the entries a_j^i are continuous, so that the determinant $\det A: U \rightarrow \mathbb{R}$ is continuous also.

By the intermediate value theorem, the continuous nowhere-vanishing function $\det A$ on the connected set U is everywhere positive or everywhere negative. Hence, $\mu = \nu$ or $\mu = -\nu$ on U . This proves that the function $f: M \rightarrow \{\pm 1\}$ is locally constant. □

Orientations and Differential Forms

Lemma 21.4 A pointwise orientation $[(X_1, \dots, X_n)]$ on a manifold M is continuous if and only if each point $p \in M$ has a coordinate neighborhood $(U, x^1, \dots, x^n)$ on which the function $(dx^1 \wedge \dots \wedge dx^n)(X_1, \dots, X_n)$ is everywhere positive.

Proof.

($\Rightarrow$) Assume that the pointwise orientation $\mu = [(X_1, \dots, X_n)]$ on M is continuous. Then every point $p \in M$ has a neighborhood W on which μ is represented by a continuous frame $(Y_1, \dots, Y_n)$. Choose a connected coordinate neighborhood $(U, x^1, \dots, x^n)$ of p contained in W . Then $Y_j = \sum_i b_j^i \frac{\partial}{\partial x^i}$ for continuous functions $[b_j^i]: U \rightarrow \mathbb{R}$. By Lemma 21.1,

$$(dx^1 \wedge \dots \wedge dx^n)(Y_1, \dots, Y_n) = \det[b_j^i] (dx^1 \wedge \dots \wedge dx^n)\left(\frac{\partial}{\partial x^1}, \dots, \frac{\partial}{\partial x^n}\right) = \det[b_j^i] \neq 0$$

We can set $(dx^1 \wedge \dots \wedge dx^n)(X_1, \dots, X_n) > 0$ on U .

Orientations and Differential Forms

Proof (continued).

($\Leftarrow$) On the chart $(U, x^1, \dots, x^n)$, suppose $X_j = \sum_i a_j^i \frac{\partial}{\partial x^i}$. As before,

$$\begin{aligned}(dx^1 \wedge \dots \wedge dx^n)(X_1, \dots, X_n) &= \det[a_j^i] (dx^1 \wedge \dots \wedge dx^n)\left(\frac{\partial}{\partial x^1}, \dots, \frac{\partial}{\partial x^n}\right) \\ &= \det[a_j^i].\end{aligned}$$

Therefore, on U , $\det[a_j^i] > 0$ and

$$[(X_1, \dots, X_n)] = [(\partial_1, \dots, \partial_n)],$$



Orientations and Differential Forms

Theorem 21.5 A manifold M of dimension n is orientable if and only if there exists a C^∞ nowhere-vanishing n -form on M .

Proof.

($\Leftarrow$) Suppose ω is a C^∞ nowhere-vanishing n -form on M . At each point $p \in M$, choose an ordered basis $(X_{1,p}, \dots, X_{n,p})$ for $T_p M$ such that $\omega_p(X_{1,p}, \dots, X_{n,p}) > 0$.

Fix $p \in M$ and let $(U, x^1, \dots, x^n)$ be a connected coordinate neighborhood of p . On U ,

$$\omega = f dx^1 \wedge \cdots \wedge dx^n$$

for a C^∞ nowhere-vanishing function f .

If $f > 0$, then on $(U, x^1, \dots, x^n)$, $(dx^1 \wedge \cdots \wedge dx^n)(X_1, \dots, X_n) > 0$.

If $f < 0$, then on $(U, -x^1, x^2, \dots, x^n)$, $(d(-x^1) \wedge dx^2 \wedge \cdots \wedge dx^n)(X_1, \dots, X_n) > 0$.

In either case, by Lemma 21.4, $\mu = [(X_1, \dots, X_n)]$ is a continuous pointwise orientation on M .

Orientations and Differential Forms

Proof (continued).

($\Rightarrow$) Suppose $[(X_1, \dots, X_n)]$ is an orientation on M . By Lemma 21.4, each point p has a coordinate neighborhood $(U, x^1, \dots, x^n)$ on which $(dx^1 \wedge \dots \wedge dx^n)(X_1, \dots, X_n) > 0$. Let $\{(U_\alpha, x_\alpha^1, \dots, x_\alpha^n)\}$ be a collection of these charts that covers M , and let $\{\rho_\alpha\}$ be a C^∞ partition of unity subordinate to the open cover $\{U_\alpha\}$. Being a locally finite sum, the n -form

$$\omega = \sum_{\alpha} \rho_{\alpha} dx_{\alpha}^1 \wedge \dots \wedge dx_{\alpha}^n$$

is well defined and C^∞ on M . Fix $p \in M$. Since $\rho_\alpha(p) \geq 0$ for all α and $\rho_\alpha(p) > 0$ for at least one α ,

$$\omega_p(X_{1,p}, \dots, X_{n,p}) = \sum_{\alpha} \rho_{\alpha}(p) (dx_{\alpha}^1 \wedge \dots \wedge dx_{\alpha}^n)_p(X_{1,p}, \dots, X_{n,p}) > 0.$$

Therefore, ω is a C^∞ nowhere-vanishing n -form on M . □

Orientations and Differential Forms

Equivalence relation between nowhere-vanishing C^∞ n -forms

If ω and ω' are two nowhere-vanishing C^∞ n -forms on a manifold M of dimension n , then $\omega = f\omega'$ for some nowhere-vanishing function f on M .

Locally, on a chart $(U, x^1, \dots, x^n)$,

$$\omega = h dx^1 \wedge \cdots \wedge dx^n \quad \text{and} \quad \omega' = g dx^1 \wedge \cdots \wedge dx^n,$$

where h and g are C^∞ nowhere-vanishing functions on U . Therefore, $f = \frac{h}{g}$ is also a C^∞ nowhere-vanishing function on U . Since U is an arbitrary chart, f is C^∞ and nowhere vanishing on M .

On a connected manifold M , such a function f is either everywhere positive or everywhere negative. In this way the nowhere-vanishing C^∞ n -forms on a connected orientable manifold M are partitioned into two equivalence classes by the equivalence relation

$$\omega \sim \omega' \iff \omega = f\omega' \text{ with } f > 0.$$

Orientations and Differential Forms

Definition

Let M be an oriented manifold with orientation $[(X_1, \dots, X_n)]$.

If ω is a C^∞ nowhere-vanishing n -form such that $\omega(X_1, \dots, X_n) > 0$, we say that ω **determines** or specifies the orientation $[(X_1, \dots, X_n)]$ and we call ω an **orientation form** on M . An oriented manifold can be described by a pair $(M, [\omega])$, where $[\omega]$ is the equivalence class of an orientation form on M .

Orientations and Differential Forms

Definition

A diffeomorphism $F: (N, [\omega_N]) \rightarrow (M, [\omega_M])$ of oriented manifolds is said to be **orientation-preserving** if $[F^*\omega_M] = [\omega_N]$; it is **orientation-reversing** if $[F^*\omega_M] = [-\omega_N]$.

Orientations and Differential Forms

Proposition 21.8 Let U and V be open subsets of $\mathbb{R}^n$, both with the standard orientation inherited from $\mathbb{R}^n$. A diffeomorphism $F: U \rightarrow V$ is orientation-preserving if and only if the Jacobian determinant $\det \left[\frac{\partial F^i}{\partial x^j} \right]$ is everywhere positive on U .

Proof.

Let $x^1, \dots, x^n$ and $y^1, \dots, y^n$ be the standard coordinates on $U \subset \mathbb{R}^n$ and $V \subset \mathbb{R}^n$. Then

$$\begin{aligned} F^*(dy^1 \wedge \cdots \wedge dy^n) &= F^*(dy^1) \wedge \cdots \wedge F^*(dy^n) = d(F^*y^1) \wedge \cdots \wedge d(F^*y^n) \\ &= dF^1 \wedge \cdots \wedge dF^n \\ &= \det \left[\frac{\partial F^i}{\partial x^j} \right] dx^1 \wedge \cdots \wedge dx^n \end{aligned}$$

Thus, $F: (U, [dx^1 \wedge \cdots \wedge dx^n]) \rightarrow (V, [dy^1 \wedge \cdots \wedge dy^n])$ is orientation-preserving if and only if $\det \left[\frac{\partial F^i}{\partial x^j} \right]$ is everywhere positive on U . □

Orientations and Atlases

Definition

An atlas on M is said to be **oriented** if for any two overlapping charts $(U, x^1, \dots, x^n)$ and $(V, y^1, \dots, y^n)$ of the atlas, the Jacobian determinant $\det \left[\frac{\partial y^i}{\partial x^j} \right]$ is everywhere positive on $U \cap V$.

: In other words, an oriented atlas is an atlas whose **transition maps are all orientation-preserving**.

Orientations and Atlases

Theorem 21.10 A manifold M is orientable if and only if it has an oriented atlas.

Proof.

($\Rightarrow$) Let $\mu = [(X_1, \dots, X_n)]$ be an orientation on the manifold M . Each point $p \in M$ has a coordinate neighborhood $(U, x^1, \dots, x^n)$ on which $(dx^1 \wedge \dots \wedge dx^n)(X_1, \dots, X_n) > 0$.

We claim that the collection $\mathcal{U} = \{(U, x^1, \dots, x^n)\}$ of these charts is an oriented atlas. If $(U, x^1, \dots, x^n)$ and $(V, y^1, \dots, y^n)$ are two overlapping charts from $\mathcal{U}$, then on $U \cap V$,

$$(dx^1 \wedge \dots \wedge dx^n)(X_1, \dots, X_n) > 0, \text{ and } (dy^1 \wedge \dots \wedge dy^n)(X_1, \dots, X_n) > 0.$$

Since

$$dy^1 \wedge \dots \wedge dy^n = \det \left[\frac{\partial y^i}{\partial x^j} \right] dx^1 \wedge \dots \wedge dx^n,$$

it follows that $\det \left[\frac{\partial y^i}{\partial x^j} \right] > 0$ on $U \cap V$. Therefore, $\mathcal{U}$ is an oriented atlas.

Orientations and Atlases

Proof (continued).

($\Leftarrow$) Suppose $\{(U, x^1, \dots, x^n)\}$ is an oriented atlas. For each $p \in U$, define μ_p to be the equivalence class of the ordered basis $\left(\frac{\partial}{\partial x^1}\Big|_p, \dots, \frac{\partial}{\partial x^n}\Big|_p\right)$ for $T_p M$.

If two charts $(U, x^1, \dots, x^n)$ and $(V, y^1, \dots, y^n)$ in the oriented atlas contain p , then by the orientability of the atlas, $\det \left[\frac{\partial y^i}{\partial x^j}\right] > 0$, so that

$$\left(\frac{\partial}{\partial x^1}\Big|_p, \dots, \frac{\partial}{\partial x^n}\Big|_p\right) \sim \left(\frac{\partial}{\partial y^1}\Big|_p, \dots, \frac{\partial}{\partial y^n}\Big|_p\right).$$

This proves that μ is a well-defined pointwise orientation on M . It is continuous because every point p has a coordinate neighborhood $(U, x^1, \dots, x^n)$ on which

$$\mu = \left[\left(\frac{\partial}{\partial x^1}, \dots, \frac{\partial}{\partial x^n}\right)\right]$$

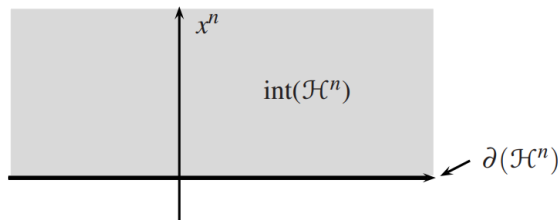
is represented by a continuous frame. □

Manifolds with Boundary

The prototype of a manifold with boundary is the closed upper half-space

$$\mathbb{H}^n = \{ (x^1, \dots, x^n) \in \mathbb{R}^n \mid x^n \geq 0 \},$$

with the subspace topology inherited from $\mathbb{R}^n$.



Manifolds with Boundary

Theorem 22.3 (Smooth invariance of domain) Let $U \subset \mathbb{R}^n$ be an open subset, $S \subset \mathbb{R}^n$ an arbitrary subset, and $f: U \rightarrow S$ a diffeomorphism. Then S is open in $\mathbb{R}^n$.

Proof.

Let $f(p)$ be an arbitrary point in S , with $p \in U$. Since $f: U \rightarrow S$ is a diffeomorphism, there are an open set $V \subset \mathbb{R}^n$ containing S and a C^∞ map $g: V \rightarrow \mathbb{R}^n$ such that $g|_S = f^{-1}$. Thus, $U \xrightarrow{f} V \xrightarrow{g} \mathbb{R}^n$ satisfies

$$g \circ f = 1_U: U \rightarrow U \subset \mathbb{R}^n,$$

the identity map on U . By the chain rule,

$$g_{*,f(p)} \circ f_{*,p} = 1_{T_p U}: T_p U \rightarrow T_p U \simeq T_p(\mathbb{R}^n).$$

Hence, $f_{*,p}$ is injective. Since U and V have the same dimension,

$$f_{*,p}: T_p U \rightarrow T_{f(p)} V$$

is invertible.

Manifolds with Boundary

Proof (Continued).

By the inverse function theorem, f is locally invertible at p . This means that there are open neighborhoods U_p of p in U and $V_{f(p)}$ of $f(p)$ in V such that

$$f: U_p \rightarrow V_{f(p)}$$

is a diffeomorphism. It follows that

$$f(p) \in V_{f(p)} = f(U_p) \subset f(U) = S.$$

Since V is open in $\mathbb{R}^n$ and $V_{f(p)}$ is open in V , the set $V_{f(p)}$ is open in $\mathbb{R}^n$. Therefore, S is open in $\mathbb{R}^n$. □

Manifolds with Boundary

Proposition 22.4 Let U and V be open subsets of the upper half-space $\mathbb{H}^n$ and $f: U \rightarrow V$ a diffeomorphism. Then f maps interior points to interior points and boundary points to boundary points.

Proof.

Let $p \in U$ be an interior point. Then p is contained in an open ball B , which is actually open in $\mathbb{R}^n$ (not just in $\mathbb{H}^n$). By smooth invariance of domain, $f(B)$ is open in $\mathbb{R}^n$ (again not just in $\mathbb{H}^n$). Therefore, $f(B) \subset (\mathbb{H}^n)^\circ$. Since $f(p) \in f(B)$, $f(p)$ is an interior point of $\mathbb{H}^n$.

If p is a boundary point in $U \cap \partial\mathbb{H}^n$, then $f^{-1}(f(p)) = p$ is a boundary point. Since $f^{-1}: V \rightarrow U$ is a diffeomorphism, by what has just been proven, $f(p)$ cannot be an interior point. Thus, $f(p)$ is a boundary point. □

Manifolds with Boundary

Definitions

- We say that a topological space M is **locally** $\mathbb{H}^n$ if every point $p \in M$ has a neighborhood U homeomorphic to an open subset of $\mathbb{H}^n$.



Fig. 22.2. Two types of open subsets of $\mathcal{H}^n$.

- A **topological n -manifold with boundary** is a second countable, Hausdorff topological space that is locally $\mathbb{H}^n$.

Manifolds with Boundary

Definitions

- Let M be a topological n -manifold with boundary. A **chart** on M is defined to be a pair (U, ϕ) consisting of an open set U in M and a homeomorphism

$$\phi: U \rightarrow \phi(U) \subset \mathbb{H}^n$$

of U with an open subset $\phi(U)$ of $\mathbb{H}^n$.

- A collection $\{(U, \phi)\}$ of charts is a C^∞ **atlas** if for any two charts (U, ϕ) and (V, ψ) , the transition map

$$\psi \circ \phi^{-1}: \phi(U \cap V) \rightarrow \psi(U \cap V) \subset \mathbb{H}^n$$

is a diffeomorphism.

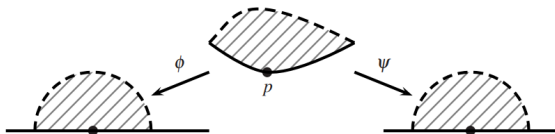
- A C^∞ **manifold with boundary** is a topological manifold with boundary together with a maximal C^∞ atlas.

Manifolds with Boundary

Definitions

Let M be a manifold of dimension n with boundary. A point p of M is called an **interior point** if in some chart (U, ϕ) , the point $\phi(p)$ is an interior point of $\mathbb{H}^n$. The set of all interior points of M is denoted by M° . Similarly, p is a **boundary point** of M if $\phi(p)$ is a boundary point of $\mathbb{H}^n$. The set of all boundary points of M is denoted by ∂M .

These concepts are well defined, independent of the charts, because if (V, ψ) is another chart, then the diffeomorphism $\psi \circ \phi^{-1} : \phi(U \cap V) \rightarrow \psi(U \cap V)$ maps $\phi(p)$ to $\psi(p)$, and so by Proposition 22.4, $\phi(p)$ and $\psi(p)$ are either both interior points or both boundary points.



Manifolds with Boundary

Example

The closed interval $[0, 1]$ is a C^∞ manifold with boundary. It has an atlas with two charts (U_1, ϕ_1) and (U_2, ϕ_2) , where $U_1 = [0, 1)$, $\phi_1(x) = x$, and $U_2 = (0, 1]$, $\phi_2(x) = 1 - x$.

However, $\{(U_1, \phi_1), (U_2, \phi_2)\}$ is not an oriented atlas, because the Jacobian determinant of the transition function

$$(\phi_2 \circ \phi_1^{-1})(x) = 1 - x$$

is negative. If we change the sign of ϕ_2 , then $\{(U_1, \phi_1), (U_2, -\phi_2)\}$ is an oriented atlas.

The Boundary of a Manifold with Boundary

Let M be a manifold of dimension n with boundary ∂M . If (U, ϕ) is a chart on M , we denote by $\phi' = \phi|_{U \cap \partial M}$ the restriction of the coordinate map ϕ to the boundary. Since ϕ maps boundary points to boundary points,

$$\phi': U \cap \partial M \rightarrow \phi(U) \cap \partial \mathbb{H}^n \subset \partial \mathbb{H}^n \simeq \mathbb{R}^{n-1}.$$

Moreover, if (U, ϕ) and (V, ψ) are two charts on M , then

$$\psi' \circ (\phi')^{-1}: \phi'(U \cap V \cap \partial M) \rightarrow \psi'(U \cap V \cap \partial M)$$

is C^∞ . Thus, an atlas $\{(U_\alpha, \phi_\alpha)\}$ for M induces an atlas

$$\{(U_\alpha \cap \partial M, \phi_\alpha|_{U_\alpha \cap \partial M})\}$$

for ∂M , making ∂M into a **manifold of dimension $n - 1$ without boundary**.

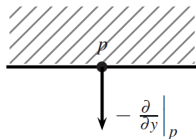
Outward-Pointing Vector Fields

Definitions

Let M be a manifold with boundary and $p \in \partial M$. We say that a tangent vector $X_p \in T_p(M)$ is **inward-pointing** if $X_p \notin T_p(\partial M)$ and there are a positive real number ε and a curve $c: [0, \varepsilon) \rightarrow M$ such that

$$c(0) = p, c((0, \varepsilon)) \subset M^\circ, \text{ and } c'(0) = X_p.$$

A vector $X_p \in T_p(M)$ is **outward-pointing** if $-X_p$ is inward-pointing. For example, on the upper half-plane $\mathbb{H}^2$, the vector $\frac{\partial}{\partial y}\big|_p$ is inward-pointing and the vector $-\frac{\partial}{\partial y}\big|_p$ is outward-pointing at a point p on the x -axis.



Outward-Pointing Vector Fields

Remark In a coordinate neighborhood $(U, \phi) = (U, x^1, \dots, x^n)$ of $p \in \partial M$, such a vector field X can be written as a linear combination

$$X_p = \sum_i a^i(p) \left. \frac{\partial}{\partial x^i} \right|_p.$$

In terms of local coordinates, a vector X_p is outward-pointing if and only if $a^n(p) < 0$.

($\Rightarrow$) There exists a curve $c : [0, \varepsilon) \rightarrow M$ such that $c(0) = p$, $c(t) \in M^\circ$ for $t > 0$, and $c'(0) = -X_p$. Since $x^n(c(0)) = 0$ and $x^n(c(t)) > 0$ ($t > 0$), we have

$$-a^n(p) = \left. \frac{d}{dt} \right|_{t=0} x^n(c(t)) \geq 0.$$

Moreover, $X_p \notin T_p(\partial M)$, so $a^n(p) \neq 0$. Hence $a^n(p) < 0$.

($\Leftarrow$) Consider

$$c(t) = \phi^{-1}(a^1(p)t, \dots, -a^n(p)t),$$

then $c(0) = p$ and $c'(0) = -X_p$.



Boundary Orientation

Proposition 22.10 On a manifold M with boundary ∂M , there is a smooth outward-pointing vector field along ∂M .

Proof.

Cover ∂M with coordinate open sets $(U_\alpha, x_\alpha^1, \dots, x_\alpha^n)$ in M .

On each U_α , the vector field $X_\alpha = -\partial/\partial x_\alpha^n$ along $U_\alpha \cap \partial M$ is smooth and outward-pointing.

Choose a partition of unity $\{\rho_\alpha\}_{\alpha \in A}$ on ∂M subordinate to the open cover $\{U_\alpha \cap \partial M\}_{\alpha \in A}$.

$$X_{\alpha,p} = \sum_{i=1}^n a^i(X_{\alpha,p}) \left. \frac{\partial}{\partial x^i} \right|_p, X := \sum_{\alpha} \rho_{\alpha} X_{\alpha}$$

$$X_p = \sum_{\alpha} \rho_{\alpha}(p) X_{\alpha,p} = \sum_{i=1}^n \left(\sum_{\alpha} \rho_{\alpha}(p) a^i(X_{\alpha,p}) \right) \left. \frac{\partial}{\partial x^i} \right|_p.$$

X is a smooth outward-pointing vector field along ∂M . □

Boundary Orientation

Proposition 22.11 Let M be an oriented n -manifold with boundary. If ω is an orientation form on M and X is a smooth outward-pointing vector field on ∂M , then $\iota_X \omega$ is a smooth nowhere-vanishing $(n-1)$ -form on ∂M . Hence, ∂M is orientable.

Proof.

Consider the following $(n-1)$ -form on ∂M . For $p \in \partial M$,

$$(\iota_X \omega)_p(v_1, \dots, v_{n-1}) = \omega_p(X_p, v_1, \dots, v_{n-1}), \quad v_1, \dots, v_{n-1} \in T_p(\partial M)$$

Since ω and X are both smooth on ∂M , so is $\iota_X \omega$.

Suppose $\iota_X \omega$ vanishes at some $p \in \partial M$. Let $e_1, \dots, e_{n-1}$ be a basis for $T_p(\partial M)$. Then $X_p, e_1, \dots, e_{n-1}$ is a basis for $T_p M$, and

$$\omega_p(X_p, e_1, \dots, e_{n-1}) = (\iota_X \omega)_p(e_1, \dots, e_{n-1}) = 0.$$

Then $\omega_p \equiv 0$ on $T_p M$, a contradiction. Therefore, $\iota_X \omega$ is nowhere vanishing on ∂M . □